

1. General Description

The EMS4798 is an audio headset analog switch that is used to detect 3.5mm accessories and switch sleeve and ring by external controller. The ground signal is routed through a pair of low-impedance ground FETs (115m Ω TYP), resulting minimal impact on audio cross-talk performance. The ground FETs of the device are designed to allow FM signal pass-through, making it possible to use the ground line of the headset as an FM antenna in mobile audio application. The EMS4798 is available in Green WLCSP- 1.17 $\times$ 1.17-9B package.

2. Features and Benefits

- Ground FET Switches R_{ON} : 110m Ω (TYP)
- High Isolation Microphone Line Switches
- Supports FM Signal Transmission Through the Ground FETs
- Reduction of Click-Pop Noise
- Power Supply Voltage Range: 2.6V to 5.0V
- Total Harmonic Distortion (MIC): 0.01% (TYP)
- Low Current Consumption: 1 μ A (TYP)
- -40 $^{\circ}$ C to +85 $^{\circ}$ C Operating Temperature Range
- Available in Green WLCSP-1.17 $\times$ 1.17-9B Package
- ESD protection:
 - HBM ESDA/JEDEC JS-001 Class 2 exceeds 2000 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 1000 V

3. Applications

- USB Type-C™, Type A and Micro B devices
- Mobile Phones, Tablets and Notebooks

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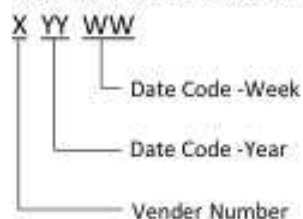
4. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS4798YFG	S4798 XYYWW	WLCSP-9B	WLCSP package, 9 bumps; 1.17 mm x 1.17 mm; 0.615 mm (Max)height	3000

MARKING INFORMATION

NOTE: XYYWW = Vendor Code and Date Code.



5.Functional Diagram

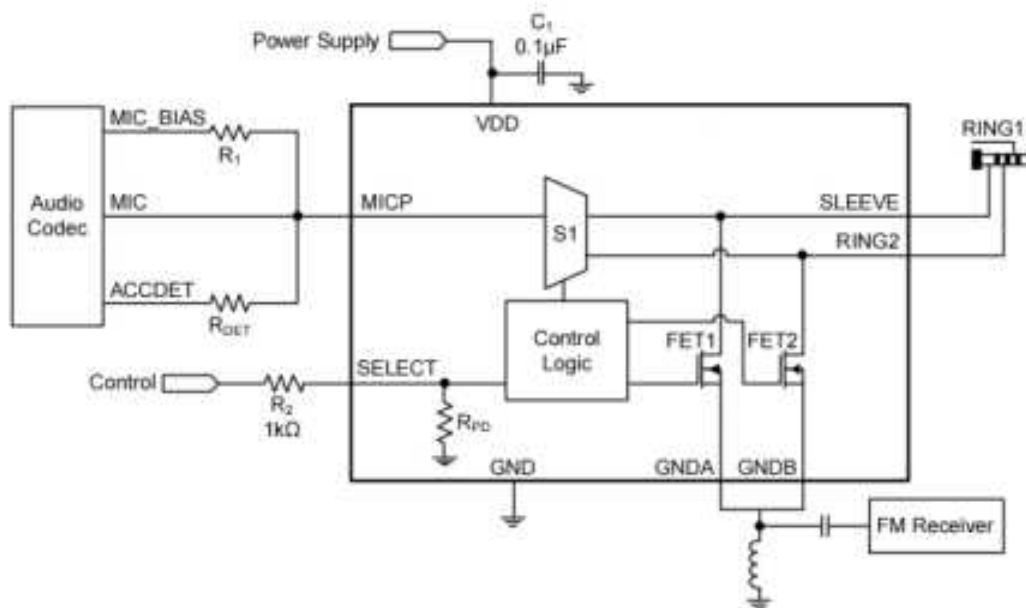


Fig 1. Typical Application

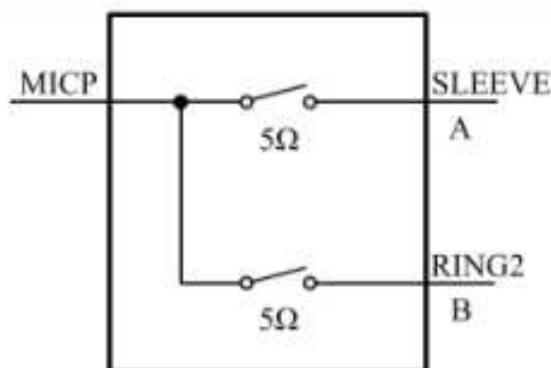


Fig 2. Functional diagram

6. Pinning Information

6.1. Pin map

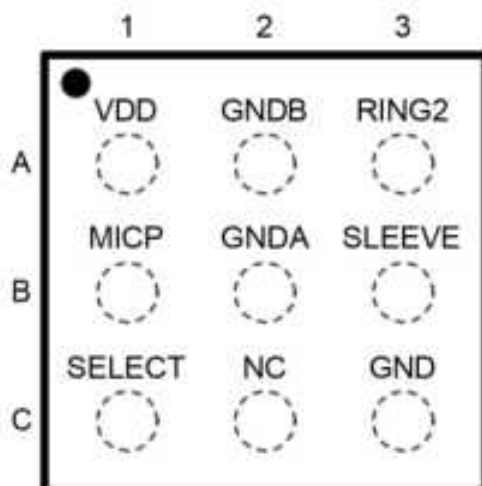


Fig 3. Top view pin configuration WLCSP-9B

6.2. Pin description

Table 2. Pin description

Symbol	Pin	Signal Type	Description
VDD	A1	-	Power Supply for the Chip
GNDB	A2	-	FET2 Ground Reference
RING2	A3	I/O	Connected to the RING2 Segment of the Jack. This pin will be routed to MICP or GNDB depending on the logic level of SELECT pin
MICP	B1	I/O	Microphone Signal Connection to Codec. Microphone bias should be fed into this pin
GNDA	B2	-	FET1 Ground Reference
SLEEVE	B3	I/O	Connected to the SLEEVE Segment of the Jack. This pin will be routed to MICP or GNDA depending on the logic level of SELECT pin
SELECT	C1	I	The Logic Signal Used to Control S1 Switch, FET1 and FET2
NC	C2	-	No Connection
GND	C3	-	Chip Ground Reference

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7.Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

SELECT	FUNCTION
L	MICP = A = SLEEVE, FET2 Turn On, FET1 Turn Off.
H	MICP = B = RING2, FET2 Turn Off, FET1 Turn On.

8.Absolute Maximum Ratings

Absolute Maximum Ratings may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions beyond those indicated in the operational sections of this specification is not implied.

Table 4. Absolute Maximum Ratings

Symbol	Parameter	Conditions	Min	Max	Unit
V _{DD}	supply voltage		-0.3	6	V
V _{IO}	voltage range on SELECT, MICP, RING2, SLEEVE		-0.3	5.5	V
R _{θJA}	thermal resistance			80	°C/W
T _{stg}	storage temperature		-55	150	°C
T _J	junction temperature under bias			125	°C
T _L	lead temperature (soldering, 10 seconds)			260	°C

9.Recommended Operating Conditions

Table 5. Recommend operating ratings

Symbol	Parameter	Conditions	Min	TYP	Max	Unit
V _{DD}	supply voltage operating		2.6		5	V
V _{IO}	voltage on SELECT, MICP, RING2, SLEEVE		0		3.3	V
T _A	operating temperature		-40	25	85	°C

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10. Electrical Characteristics

Table 6. Electrical characteristics

$V_{DD} = 2.6V$ to $5.0V$, typical values are at $V_{DD} = 3.3V$, $T_A = +25^\circ C$, unless otherwise noted.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
POWER SUPPLY						
V _{DD}	supply voltage range		2.6		5	V
I _Q	quiescent Current	V _{DD} = 4.5V, V _{MICP} = 1.8V to V _{DD} , SELECT = Low or SELECT = High (after detection)		0.1	1	uA
V _{IO}	input/output voltage range	V _{DD} ≤ 3.3 V	0		V _{DD}	V
		V _{DD} ≥ 3.3 V	0		3.3	V
SEL/_EN DIGITAL INPUT CONTROL						
V _{IH}	input logic high for SELECT	V _{DD} = 2.6V	1.3		V _{DD}	V
		V _{DD} = 3.3V	1.4		V _{DD}	
		V _{DD} = 4.5V	1.5		V _{DD}	
V _{IL}	input logic low for SELECT	V _{DD} = 2.6V	0		0.3	V
		V _{DD} = 3.3V	0		0.4	
		V _{DD} = 4.5V	0		0.5	
SWITCH ON RESISTANCE AND LEAKAGE						
R _{PD}	on-resistance			460		kΩ
R _{F1}	FET1 On Resistance	V _{DD} = 2.6V, V _{GND} = 0V, I _{GND} = 10mA		110		mΩ
R _{F2}	FET2 On Resistance			110		
R _{S1A}	S1 On Resistance (Closed to SLEEVE)	V _{DD} = 2.6V, V _{SLEEVE/RING2} = 0 V to 2.6V, I _{MIC} = ±10mA		5.3		Ω
R _{S1B}	S1 On Resistance (Closed to EING2)			5.3		Ω
I _{LFET(OFF)}	FET1,FET2, OFF leakage current	V _{DD} = 3.3V, V _{SELECT} = 0V or 3.3V; V _{IN} = 0V to 3.3V, V _O = 0V			1	uA
I _{S1AB(OFF)}	S1A, S1B, OFF leakage current				1	uA

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Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$I_{S1AB(ON)}$	S1A,S1B ON leakage current				1	μA
SWITCH DYNAMICS						
t_{BBM}	break-before-make time	$V_{IS} = 0.4 V, R_L = 50 \Omega$		1.4		μs
t_{ON}	turn-on time	$V_{IS} = 0.4 V, R_L = 50 \Omega$ SELECT switches from High to Low		1.4		μs
t_{OFF}	turn-off time	$V_{IS} = 0.4 V, R_L = 50 \Omega$ SELECT switches from Low to High		70		ns

[1] Flatness is defined as the difference between maximum and minimum value of ON-resistance at the specified analog signal voltage points.

[2] RON matching between channels is calculated by subtracting the channel with the lowest max Ron value from the channel with the highest max RON value.

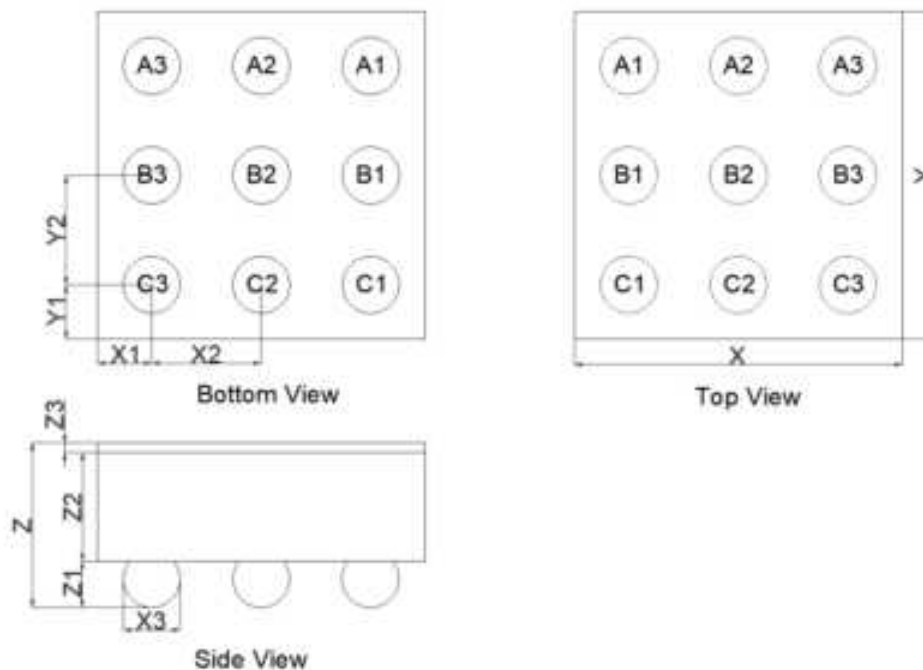
[3] Crosstalk is inversely proportional to source impedance.

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11. Package Outline

WLCSP-9B



Symbol	Dimensions In Millimeter		
	Min.	Typ.	Max.
X	1.145	1.17	1.195
Y	1.145	1.17	1.195
X1	-	0.185	-
X2	-	0.400	-
X3	0.245	0.270	0.295
Y1	-	0.185	-
Y2	-	0.400	-
Z	0.535	0.575	0.615
Z1	0.170	0.195	0.22
Z2	0.345	0.355	0.365
Z3	0.02	0.025	0.03

12. Tape and Reel Information

13.1. Carrier tape dimensions

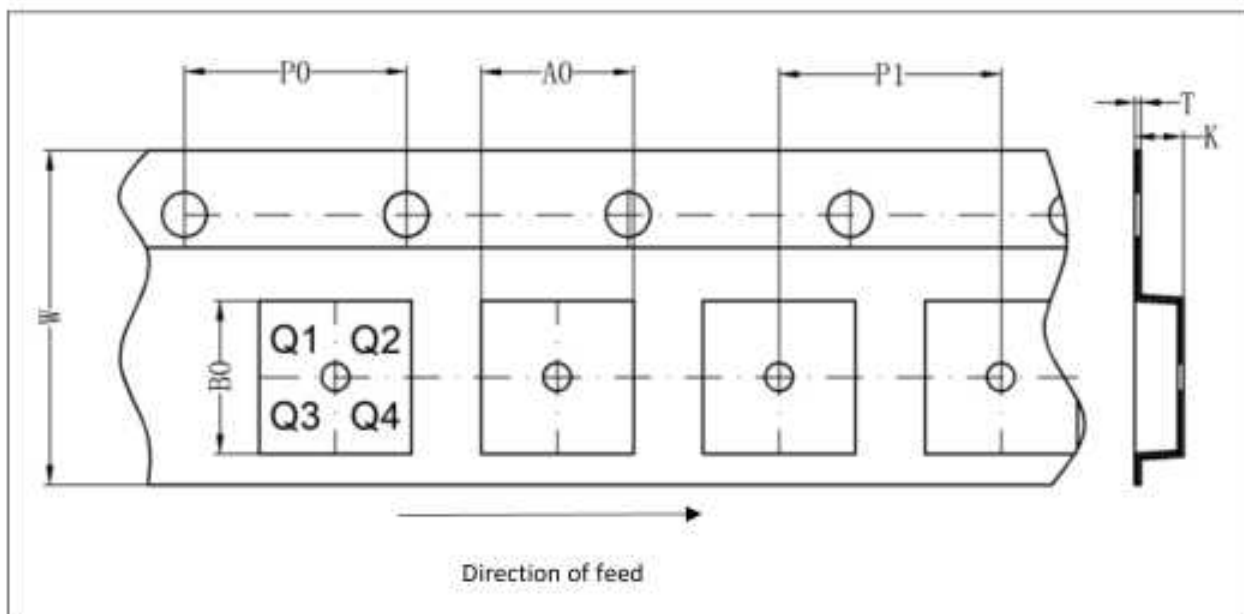


Table 7. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
WLCSP-9B								

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13.2. Reel and box dimensions

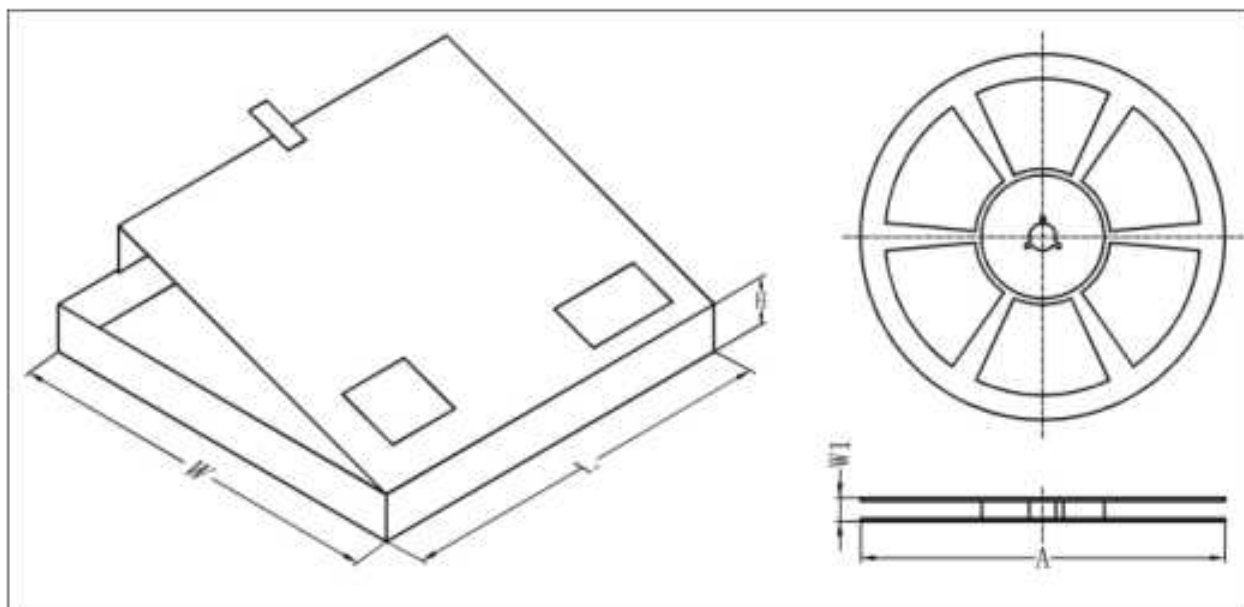


Table 8. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
WLCSP-9B	YFG					

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

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11. Abbreviations

Table 9. Abbreviations

Acronym	Description
ESD	ElectroStatic Discharge
HBM	Human Body Model

12. Revision History

Table 10. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS4798 Rev. 1.0	Apr 08, 2026	Product datasheet		